

PRECISION NDE SERVICES INC.

ULTRASONIC THICKNESS SURVEY FINAL INSPECTION REPORT

Report Number:	PNDE-GLCP-2024-0847
Revision:	Rev 0
Date Issued:	December 15, 2024
Client:	Great Lakes Chemical Processing Inc.
Facility:	Sarnia Refinery Complex — Hydroprocessing Unit
Unit:	Naphtha Hydrotreater (NHT-100)
Survey Period:	November 4 – November 22, 2024

APPROVAL

Role	Name	Certification	Signature	Date
Prepared By	M. Rajendran	CGSB UT Level II #UT-2019-4472	_____	2024-12-10
Reviewed By	S. Petrov	CGSB UT Level III #UT-2015-2108	_____	2024-12-12
Approved By	D. Okonkwo	API 570 Inspector #570-0847	_____	2024-12-15

TABLE OF CONTENTS

1. Executive Summary.....

3

2. Scope of Work.....

3

3. Reference Documents.....

4

4. Personnel & Qualifications.....

4

5. Equipment & Calibration.....

5

6. Inspection Results — Data Tables.....

6

7. Trending & Corrosion Analysis.....

9

8. Findings & Dispositions.....

11

9. Recommendations.....

13

10. Appendix A — Daily Calibration Verification Records.....

14

11. Appendix B — Photograph Log.....

15

12. Appendix C — Personnel Certification Records.....

16

13. Appendix D — Equipment Calibration Certificates.....

16

14. Appendix E — Isometric Drawings with CML Locations.....

17

15. Appendix F — Raw Data (Gauge Data Exports).....

17

1. EXECUTIVE SUMMARY

Precision NDE Services Inc. (PNDE) was contracted by Great Lakes Chemical Processing Inc. (GLCP) to perform an ultrasonic thickness (UT) survey of piping systems within the Naphtha Hydrotreater unit (NHT-100) at the Sarnia Refinery Complex. The survey was conducted between November 4 and November 22, 2024, encompassing 14 process lines and 87 Condition Monitoring Locations (CMLs).

Of the 87 CMLs surveyed, 82 were fully accessible and measured. Five CMLs were inaccessible due to insulation and scaffolding constraints (documented in Section 8). A total of 328 individual thickness readings were recorded using the minimum four-point (90-degree interval) measurement protocol.

Key findings include **2 CRITICAL locations** (readings below t_{min}), **4 Alert locations** (readings below 110% of t_{min}), and **6 Monitor locations** (short-term corrosion rates exceeding 10 mpy). Immediate engineering review is recommended for the two CRITICAL CMLs. The owner/operator was notified of CRITICAL findings on November 12, 2024, via GLCP Notification Form NF-2024-0093.

Overall, the NHT-100 piping system remains within acceptable service limits at 93% of surveyed CMLs. The average short-term corrosion rate across all lines is 4.8 mpy, consistent with the expected corrosion environment for naphtha hydrotreating service.

2. SCOPE OF WORK

The scope of this inspection encompassed the complete UT thickness survey of the NHT-100 Naphtha Hydrotreater piping circuit at the GLCP Sarnia Refinery Complex, as defined in GLCP Purchase Order PO-2024-7821 and Work Scope Document WSD-NHT100-2024-Rev2.

Parameter	Detail
Facility	GLCP Sarnia Refinery Complex
Unit	Naphtha Hydrotreater (NHT-100)
Total Lines Surveyed	14 piping lines
Total CMLs	87 (82 measured, 5 inaccessible)
Total Readings	328 individual thickness readings
Survey Start Date	November 4, 2024
Survey End Date	November 22, 2024
Working Days	15 (including 1 weather delay)
Previous Survey	October 2019 (Report PNDE-GLCP-2019-0412)
Inspection Method	Contact UT (pulse-echo, straight beam)

3. REFERENCE DOCUMENTS

Document	Revision / Edition
API 570 — Piping Inspection Code	4th Edition, February 2016
API 574 — Inspection Practices for Piping System Components	4th Edition, November 2016
ASME B31.3 — Process Piping	2022 Edition
ASME Section V, Article 5 — UT Examination Methods for Materials	2023 Edition
ASNT SNT-TC-1A — Personnel Qualification and Certification	2020 Edition
GLCP-NDT-001 — General NDT Procedure for Piping Thickness Surveys	Rev 3, January 2023
GLCP-QAP-005 — Quality Assurance Program for Contractor NDE	Rev 2, March 2022
PNDE-UT-SOP-010 — UT Thickness Measurement Standard Operating Procedure	Rev 5, June 2024
GLCP Piping Specification GS-P-101 (NHT-100 Carbon Steel Piping)	Rev 4, August 2021
Previous Inspection Report PNDE-GLCP-2019-0412	Rev 0, December 2019

4. PERSONNEL & QUALIFICATIONS

4.1 Inspector Certification Summary

All inspection personnel assigned to this survey meet or exceed the minimum qualification requirements specified in GLCP-NDT-001 §3.1 and §3.2. Certification is in accordance with ASNT SNT-TC-1A and CGSB 48.9712. Full certification records are appended in Appendix C.

Name	Role	Cert Level	Cert Number	Expiry	Years Exp.
M. Rajendran	Lead Inspector	CGSB UT Level II	UT-2019-4472	2025-07-15	12
K. Nakamura	Inspector	CGSB UT Level II	UT-2020-5891	2026-03-22	8
A. Thompson	Inspector	CGSB UT Level II	UT-2021-6234	2026-09-01	6
S. Petrov	QA Reviewer	CGSB UT Level III	UT-2015-2108	2025-11-30	18

4.2 Minimum Qualification Confirmation

All inspectors assigned to this project hold a minimum UT Level II certification with at least 5 years of field experience in piping thickness surveys, in compliance with GLCP-NDT-001 §3.2. Verification of current certification status was confirmed prior to survey commencement.

4.3 Site-Specific Safety Orientation

All PNDE personnel completed the GLCP site-specific safety orientation (Course ID: GLCP-SO-2024-NHT) on November 3, 2024, prior to survey commencement. Orientation covered H₂S awareness, confined space protocols, hot work permits, and emergency evacuation procedures. Completion records are maintained in the GLCP contractor safety database (Ref: CSO-2024-0847-01 through -04).

5. EQUIPMENT & CALIBRATION

5.1 Equipment Identification

Item	Description	Model	Serial Number	Cal. Due
Gauge 1	UT Thickness Gauge	Olympus 38DL PLUS	SN-38DL-2022-08741	2025-04-15
Gauge 2	UT Thickness Gauge	Olympus 38DL PLUS	SN-38DL-2023-11205	2025-08-22
Probe 1	D797-SM, 5 MHz, 0.25" dia.	Olympus D797-SM	SN-D797-2023-4521	N/A
Probe 2	D797-SM, 5 MHz, 0.25" dia.	Olympus D797-SM	SN-D797-2024-0087	N/A
Cal Block	4-Step Cal Block, CS	PNDE Custom (ASTM E797)	SN-CB-2021-0034	2025-06-10

5.2 Calibration Certificates

All gauges are calibrated by an ISO/IEC 17025 accredited laboratory with traceability to NIST standards. Current calibration certificates for all equipment are appended in Appendix D. Calibration block certificate (SN-CB-2021-0034) is traceable to NRC reference standards.

5.3 Daily Calibration Verification

Daily calibration verification was performed in accordance with GLCP-NDT-001 §4.3 at the start and end of each inspection shift. All verification readings were within the required ± 0.002 " tolerance. Complete daily verification records including date, time, operator ID, and readings are provided in Appendix A. A summary of daily calibration status is shown below.

Table 5.3a — Daily Calibration Verification Summary (first 3 days shown; complete records in Appendix A)

Date	Gauge S/N	Operator	AM Check (in.)	PM Check (in.)	Tolerance	Status
2024-11-04	08741	MR	0.200 / 0.400 / 0.600	0.200 / 0.401 / 0.600	± 0.002 "	PASS
2024-11-05	08741	MR	0.200 / 0.400 / 0.600	0.200 / 0.400 / 0.601	± 0.002 "	PASS
2024-11-05	11205	KN	0.200 / 0.400 / 0.600	0.201 / 0.400 / 0.600	± 0.002 "	PASS
2024-11-06	08741	MR	0.200 / 0.400 / 0.600	0.200 / 0.400 / 0.600	± 0.002 "	PASS
2024-11-06	11205	KN	0.201 / 0.400 / 0.600	0.200 / 0.401 / 0.600	± 0.002 "	PASS

5.4 Material Velocity Settings

Sound velocity was set to 0.2330 in/ μ s (5,918 m/s) for carbon steel piping, consistent with GLCP piping specification GS-P-101 (SA-106 Gr. B / SA-234 WPB material). Velocity was verified against the known calibration block step thicknesses at the start of each survey day.

6. INSPECTION RESULTS — DATA TABLES

Results are tabulated by piping line and CML location in accordance with GLCP-NDT-001 §7.3. Each CML was measured using a minimum of four readings at 90-degree intervals around the pipe circumference. The minimum reading at each CML is reported as the governing thickness. Previous survey data (October 2019) is included for trending purposes.

6.1 Line NHT-L-101 — Naphtha Feed Line

Line ID	CML	Component	NPS (in.)	Nom. (in.)	t-min (in.)	2019 (in.)	2024 Min (in.)	Date	Operator	Equip. S/N
NHT-L-101	001	Str. Pipe	6	0.280	0.189	0.271	0.262	2024-11-04	MR	08741
NHT-L-101	002	Elbow	6	0.280	0.189	0.264	0.251	2024-11-04	MR	08741
NHT-L-101	003	Tee	6	0.280	0.189	0.268	0.258	2024-11-04	MR	08741
NHT-L-101	004	Str. Pipe	6	0.280	0.189	0.275	0.268	2024-11-04	MR	08741
NHT-L-101	005	Reducer	6×4	0.280	0.189	0.260	0.243	2024-11-05	MR	08741
NHT-L-101	006	Elbow	4	0.237	0.154	0.228	0.215	2024-11-05	MR	08741
NHT-L-101	007	Str. Pipe	4	0.237	0.154	0.231	0.224	2024-11-05	MR	08741

6.2 Line NHT-L-102 — Reactor Effluent Line

Line ID	CML	Component	NPS (in.)	Nom. (in.)	t-min (in.)	2019 (in.)	2024 Min (in.)	Date	Operator	Equip. S/N
NHT-L-102	008	Str. Pipe	8	0.322	0.218	0.310	0.296	2024-11-06	KN	11205
NHT-L-102	009	Elbow	8	0.322	0.218	0.298	0.271	2024-11-06	KN	11205
NHT-L-102	010	Elbow	8	0.322	0.218	0.285	0.237	2024-11-07	KN	11205
NHT-L-102	011	Tee	8	0.322	0.218	0.275	0.211	2024-11-07	KN	11205
NHT-L-102	012	Str. Pipe	8	0.322	0.218	0.305	0.288	2024-11-07	KN	11205
NHT-L-102	013	Elbow	8	0.322	0.218	0.290	0.249	2024-11-08	KN	11205
NHT-L-102	014	Reducer	8×6	0.322	0.218	0.280	0.240	2024-11-08	KN	11205
NHT-L-102	015	Str. Pipe	6	0.280	0.189	0.270	0.254	2024-11-08	KN	11205

6.3 Line NHT-L-103 — Stripper Overhead Line

Line ID	CML	Component	NPS (in.)	Nom. (in.)	t-min (in.)	2019 (in.)	2024 Min (in.)	Date	Operator	Equip. S/N
NHT-L-103	016	Str. Pipe	10	0.365	0.250	0.358	0.351	2024-11-11	AT	08741
NHT-L-103	017	Elbow	10	0.365	0.250	0.350	0.340	2024-11-11	AT	08741

Line ID	CML	Component	NPS (in.)	Nom. (in.)	t-min (in.)	2019 (in.)	2024 Min (in.)	Date	Oper .	Equip. S/N
NHT-L-103	018	Tee	10	0.365	0.250	0.345	0.332	2024-11-11	AT	08741
NHT-L-103	019	Str. Pipe	10	0.365	0.250	0.355	0.348	2024-11-11	AT	08741
NHT-L-103	020	Elbow	10	0.365	0.250	0.340	0.325	2024-11-12	AT	08741
NHT-L-103	021	Str. Pipe	10	0.365	0.250	0.352	0.344	2024-11-12	AT	08741

6.4 Line NHT-L-104 — Hydrogen Makeup Line

Line ID	CML	Component	NPS (in.)	Nom. (in.)	t-min (in.)	2019 (in.)	2024 Min (in.)	Date	Oper .	Equip. S/N
NHT-L-104	022	Str. Pipe	3	0.216	0.138	0.205	0.190	2024-11-13	MR	08741
NHT-L-104	023	Elbow	3	0.216	0.138	0.198	0.172	2024-11-13	MR	08741
NHT-L-104	024	Elbow	3	0.216	0.138	0.185	0.134	2024-11-13	MR	08741
NHT-L-104	025	Tee	3	0.216	0.138	0.195	0.165	2024-11-13	MR	08741
NHT-L-104	026	Str. Pipe	3	0.216	0.138	0.208	0.198	2024-11-14	MR	08741
NHT-L-104	027	Elbow	3	0.216	0.138	0.190	0.155	2024-11-14	MR	08741

Note: Detailed data tables for all 14 lines (87 CMLs) are provided in Appendix F — Raw Data. Lines NHT-L-105 through NHT-L-114 data tables follow the same format and are included in the complete dataset appendix. Only lines with notable findings are shown in this section.

7. TRENDING & CORROSION ANALYSIS

7.1 Historical Trending Comparison

Thickness measurements from the current survey (November 2024) are compared against the previous survey (October 2019) to establish short-term corrosion rates. The inspection interval is approximately 5.1 years (1,862 days). Long-term corrosion rates incorporate the original installation thickness data from 2008 commissioning records where available.

7.2 Corrosion Rate & Remaining Life Summary

Table 7.2 — Corrosion Rate and Remaining Life Summary (CMLs with notable findings and representative acceptable CMLs)

Line ID	CML	Component	2019 (in.)	2024 (in.)	Loss (in.)	ST Rate (mpy)	LT Rate (mpy)	Rem. Life (yrs)	Next Insp. Date	Category
NHT-L-10 1	001	Str. Pipe	0.271	0.262	0.009	1.8	1.5	40.6	2030-11	Acceptable
NHT-L-10 1	002	Elbow	0.264	0.251	0.013	2.5	2.2	24.8	2029-11	Acceptable
NHT-L-10 1	003	Tee	0.268	0.258	0.010	2.0	1.7	34.5	2030-11	Acceptable
NHT-L-10 1	005	Reducer	0.260	0.243	0.017	3.3	3.0	16.4	2029-11	Acceptable
NHT-L-10 2	009	Elbow	0.298	0.271	0.027	5.3	4.8	10.0	2027-11	Acceptable
NHT-L-10 2	010	Elbow	0.285	0.237	0.048	9.4	7.1	2.0	2025-11	Alert
NHT-L-10 2	011	Tee	0.275	0.211	0.064	12.5	9.2	-0.6	IMMED.	CRITICAL
NHT-L-10 2	013	Elbow	0.290	0.249	0.041	8.0	6.5	3.9	2026-11	Alert
NHT-L-10 2	014	Reducer	0.280	0.240	0.040	7.8	6.2	2.8	2026-05	Alert
NHT-L-10 3	017	Elbow	0.350	0.340	0.010	2.0	1.8	45.0	2030-11	Acceptable
NHT-L-10 3	020	Elbow	0.340	0.325	0.015	2.9	2.5	25.9	2029-11	Acceptable
NHT-L-10 4	023	Elbow	0.198	0.172	0.026	5.1	4.5	6.7	2027-05	Monitor
NHT-L-10 4	024	Elbow	0.185	0.134	0.051	10.0	8.5	-0.4	IMMED.	CRITICAL
NHT-L-10 4	025	Tee	0.195	0.165	0.030	5.9	5.2	5.2	2027-05	Monitor
NHT-L-10 4	027	Elbow	0.190	0.155	0.035	6.9	5.8	2.5	2026-05	Alert
NHT-L-10 6	038	Elbow	0.255	0.198	0.057	11.2	8.1	3.5	2026-05	Monitor

Line ID	CML	Component	2019 (in.)	2024 (in.)	Loss (in.)	ST Rate (mpy)	LT Rate (mpy)	Rem. Life (yrs)	Next Insp. Date	Category
NHT-L-106	039	Tee	0.248	0.190	0.058	11.4	8.3	2.9	2025-11	Monitor
NHT-L-108	052	Elbow	0.312	0.260	0.052	10.2	7.5	4.1	2026-11	Monitor
NHT-L-108	053	Str. Pipe	0.318	0.272	0.046	9.0	6.8	6.0	2027-05	Monitor

Corrosion Rate Calculation Method (per API 570 §7.1):

Short-term corrosion rate (mpy) = (Previous thickness – Current thickness) / Interval years × 1000

Long-term corrosion rate (mpy) = (Nominal thickness – Current thickness) / Service years × 1000

Remaining life (years) = (Current thickness – t_{min}) / Short-term corrosion rate

Next recommended inspection date is calculated per API 570 §6, using one-half of the remaining life or a maximum interval of 5 years for general corrosion, whichever is shorter.

7.3 Corrosion Rate Distribution

Across all 82 measured CMLs, the short-term corrosion rate distribution is as follows: 58 CMLs (71%) exhibit rates below 5 mpy; 18 CMLs (22%) between 5 and 10 mpy; and 6 CMLs (7%) above 10 mpy. The overall average short-term rate is 4.8 mpy. Lines NHT-L-102, NHT-L-104, NHT-L-106, and NHT-L-108 show the highest corrosion activity, consistent with the higher-temperature reactor effluent and hydrogen service environments.

8. FINDINGS & DISPOSITIONS

8.1 Findings Summary by Category

All findings are categorized in accordance with GLCP-NDT-001 §6.2. The severity classification thresholds are defined as follows:

Category	Criteria	Count
CRITICAL	Measured thickness < t-min	2
Alert	Measured thickness < 110% of t-min	4
Monitor	Short-term corrosion rate > 10 mpy	6
Acceptable	All other CMLs within limits	70
Inaccessible	Unable to measure — documented	5

8.2 CRITICAL Findings — Below t-min

■ IMMEDIATE ENGINEERING REVIEW REQUIRED

Two CMLs were found to have measured thicknesses below the minimum required wall thickness (t_{min}) calculated per ASME B31.3. The owner/operator was notified immediately in accordance with GLCP-NDT-001 §6.3. Notification was documented via GLCP Notification Form NF-2024-0093, transmitted to the GLCP Operations Manager and Mechanical Integrity Engineer on November 12, 2024.

CML	Line	Component	t-min (in.)	Measured (in.)	Deficit	Rate (mpy)
011	NHT-L-102	Tee (8")	0.218	0.211	0.007 below	12.5
024	NHT-L-104	Elbow (3")	0.138	0.134	0.004 below	10.0

Disposition — CML-011 (NHT-L-102): Fitness-for-Service (FFS) evaluation per API 579-1 recommended. Restrict operating pressure to 85% of MAWP pending engineering assessment. Consider immediate replacement of tee component during next available maintenance window.

Disposition — CML-024 (NHT-L-104): Immediate replacement of elbow recommended. Given the 3" hydrogen makeup service and the accelerated corrosion rate, this component should be replaced with upgraded metallurgy (consider 1.25 Cr-0.5 Mo per process engineering review). Temporary repair clamp authorized per API 570 pending replacement.

8.3 Alert Findings — Below 110% of t-min

CML	Line	Component	t-min (in.)	110% t-min (in.)	Measured (in.)	Rate (mpy)
010	NHT-L-102	Elbow (8")	0.218	0.240	0.237	9.4

CML	Line	Component	t-min (in.)	110% t-min (in.)	Measured (in.)	Rate (mpy)
013	NHT-L-102	Elbow (8")	0.218	0.240	0.249	8.0
014	NHT-L-102	Reducer (8×6")	0.218	0.240	0.240	7.8
027	NHT-L-104	Elbow (3")	0.138	0.152	0.155	6.9

Disposition — Alert CMLs: Increased monitoring frequency recommended (6-month re-inspection). Engineering review to determine if accelerated corrosion warrants metallurgy upgrade or process chemistry adjustment. FFS evaluation per API 579-1 Part 4 (general metal loss) is recommended for CMLs 010 and 014 given their proximity to t_{min} .

8.4 Monitor Findings — Corrosion Rates > 10 mpy

Note: CMLs 011 and 024 appear in both CRITICAL and Monitor categories.

CML	Line	Component	Measured (in.)	ST Rate (mpy)	Rem. Life (yrs)
011	NHT-L-102	Tee (8")	0.211	12.5	-0.6
024	NHT-L-104	Elbow (3")	0.134	10.0	-0.4
038	NHT-L-106	Elbow (6")	0.198	11.2	3.5
039	NHT-L-106	Tee (6")	0.190	11.4	2.9
052	NHT-L-108	Elbow (8")	0.260	10.2	4.1
053	NHT-L-108	Str. Pipe (8")	0.272	9.0	6.0

8.5 Inaccessible CMLs

Five CMLs could not be measured during this survey. The reasons for inaccessibility are documented below per GLCP-NDT-001 §7.2. Photographs of the inaccessible locations are included in Appendix B.

CML	Line	Component	Reason for Inaccessibility
034	NHT-L-105	Elbow (6")	Encapsulated in calcium silicate insulation; insulation removal not authorized
047	NHT-L-107	Str. Pipe (10")	Located above pipe rack Tier 3; scaffolding not available during survey window
048	NHT-L-107	Elbow (10")	Adjacent to active steam trace — surface temperature exceeded safe contact limit
065	NHT-L-110	Tee (4")	Fireproofing applied to structural support obstructs CML access point
071	NHT-L-111	Str. Pipe (6")	CML location buried within pipe support clamp; requires disassembly

8.6 Anomalies

The following anomalies were noted during the inspection and are reported for reference:

CML-009 (NHT-L-102): Heavy internal scale buildup indicated by inconsistent back-wall echo. Readings were repeatable within ± 0.003 " after surface preparation, but the presence of internal deposits may affect measurement accuracy. Recommend internal visual inspection at next opportunity.

CML-038 (NHT-L-106): External surface exhibited significant pitting corrosion under insulation (CUI). Localized pitting depths estimated at 0.030"–0.050" observed during surface preparation. Area photographed and marked for CUI investigation program.

CML-052 (NHT-L-108): Unexpected low readings at the 6 o'clock position suggest possible localized erosion-corrosion at the elbow extrados. Pattern is consistent with multiphase flow impingement. Recommend supplemental PAUT scan to map wall loss profile.

9. RECOMMENDATIONS

9.1 Immediate Actions

9.1.1 CML-011 (NHT-L-102, 8" Tee): Commission FFS evaluation per API 579-1 Part 4. Restrict operating pressure to 85% MAWP pending assessment. Plan replacement during next scheduled turnaround (recommend SA-335 P11 or equivalent for improved corrosion resistance).

9.1.2 CML-024 (NHT-L-104, 3" Elbow): Replace elbow immediately. Authorize temporary repair clamp per API 570 pending replacement. Review metallurgy selection with process engineering — 1.25Cr-0.5Mo (SA-335 P11) recommended for hydrogen makeup service.

9.2 Re-Inspection Intervals

Re-inspection intervals are established per API 570 §6 based on the calculated remaining life at each CML. The governing principle is one-half of remaining life, with a maximum interval of 5 years and a minimum of 6 months for CMLs approaching alert thresholds.

Priority	CMLs	Recommended Interval	Next Inspection
CRITICAL	011, 024	Immediate engineering review; replace/repair	Immediate
Alert	010, 013, 014, 027	6-month re-inspection	May 2025
Monitor	038, 039, 052, 053	12-month re-inspection	November 2025
Acceptable	All remaining (70 CMLs)	5-year interval (next general survey)	November 2029
Inaccessible	034, 047, 048, 065, 071	Measure at next turnaround or when access permits	Next T/A

9.3 Engineering Review Referrals

9.3.1 FFS Evaluation — API 579-1: CMLs 010, 011, 014, and 024 are referred for Fitness-for-Service evaluation. CMLs 011 and 024 (below t_{min}) require Level 2 assessment minimum. CMLs 010 and 014 (approaching t_{min}) should be assessed at Level 1.

9.3.2 Metallurgy Review: Lines NHT-L-102 and NHT-L-104 exhibit significantly higher corrosion rates than the remainder of the NHT-100 circuit. Process engineering review is recommended to evaluate whether process conditions have changed (e.g., H_2S concentration, reactor effluent temperature) and whether metallurgy upgrades are warranted.

9.4 Process Review Recommendations

9.4.1 The accelerated corrosion observed in lines NHT-L-102 (reactor effluent) and NHT-L-106 suggests a potential increase in corrosive species (ammonium bisulfide) concentration. PNDE recommends that GLCP process engineering review the following: (a) reactor outlet temperature trending; (b) wash water injection rates and distribution; (c) overhead system pH monitoring data; and (d) crude/feed slate changes in the preceding 18 months.

9.4.2 The CUI finding at CML-038 (NHT-L-106) warrants initiation of a broader CUI inspection program for insulated carbon steel piping operating between 25°F and 350°F in the NHT-100 unit, consistent with API 570 Appendix C guidelines.

APPENDIX A — DAILY CALIBRATION VERIFICATION RECORDS

Complete daily calibration verification records per GLCP-NDT-001 §4.3. Each record includes date, time, operator identification, gauge serial number, step block readings, and pass/fail determination. Tolerance: ± 0.002 " on all calibration steps.

Date	Time	Gauge S/N	Oper.	Step 1 (0.200")	Step 2 (0.400")	Step 3 (0.600")	Status
2024-11-04	06:45	08741	MR	0.200	0.400	0.600	PASS
2024-11-04	16:30	08741	MR	0.200	0.401	0.600	PASS
2024-11-05	06:50	08741	MR	0.200	0.400	0.600	PASS
2024-11-05	16:25	08741	MR	0.200	0.400	0.601	PASS
2024-11-05	06:55	11205	KN	0.200	0.400	0.600	PASS
2024-11-05	16:35	11205	KN	0.201	0.400	0.600	PASS
2024-11-06	06:48	08741	MR	0.200	0.400	0.600	PASS
2024-11-06	16:28	08741	MR	0.200	0.400	0.600	PASS
2024-11-06	06:52	11205	KN	0.201	0.400	0.600	PASS
2024-11-06	16:30	11205	KN	0.200	0.401	0.600	PASS
2024-11-07	06:45	11205	KN	0.200	0.400	0.600	PASS
2024-11-07	16:32	11205	KN	0.200	0.400	0.601	PASS
2024-11-08	06:50	11205	KN	0.200	0.400	0.600	PASS
2024-11-08	16:28	11205	KN	0.201	0.400	0.600	PASS
2024-11-11	06:48	08741	AT	0.200	0.400	0.600	PASS
2024-11-11	16:35	08741	AT	0.200	0.401	0.600	PASS
2024-11-12	06:45	08741	AT	0.200	0.400	0.600	PASS
2024-11-12	16:30	08741	AT	0.200	0.400	0.600	PASS
2024-11-13	06:50	08741	MR	0.200	0.400	0.600	PASS
2024-11-13	16:32	08741	MR	0.200	0.400	0.601	PASS
2024-11-14	06:48	08741	MR	0.200	0.400	0.600	PASS
2024-11-14	16:28	08741	MR	0.201	0.400	0.600	PASS

Note: Records continue for all working days through November 22, 2024. All verifications passed within ± 0.002 " tolerance. Complete records maintained in PNDE QA File Ref: QAF-2024-0847-CAL.

APPENDIX B — PHOTOGRAPH LOG

Photographs are organized by category per GLCP-NDT-001 §7.2. All photographs are geotagged and timestamped. High-resolution originals are archived in PNDE digital records system (Ref: IMG-GLCP-2024-0847). Representative photo log entries are listed below.

Photo ID	CML / Location	Category	Description	Date	File Ref
P-001	CML-001	CML Location	NHT-L-101 CML-001 straight pipe, measurement location marked	2024-11-04	IMG-0847-001
P-002	CML-002	CML Location	NHT-L-101 CML-002 elbow, 6" 90° LR	2024-11-04	IMG-0847-002
P-010	CML-011	CML Location	NHT-L-102 CML-011 tee, 8" — CRITICAL location	2024-11-07	IMG-0847-010
P-011	CML-011	Gauge Display	Gauge reading 0.211" — below t-min (0.218")	2024-11-07	IMG-0847-011
P-012	CML-024	CML Location	NHT-L-104 CML-024 elbow, 3" — CRITICAL location	2024-11-13	IMG-0847-024
P-013	CML-024	Gauge Display	Gauge reading 0.134" — below t-min (0.138")	2024-11-13	IMG-0847-025
P-014	CML-010	Gauge Display	Gauge reading 0.237" — Alert threshold	2024-11-07	IMG-0847-012
P-020	CML-038	Surface Cond.	External pitting under insulation at CML-038	2024-11-15	IMG-0847-038
P-021	CML-052	Surface Cond.	Elbow extrados showing localized wall loss pattern	2024-11-18	IMG-0847-052
P-030	CML-034	Inaccessible	Insulation encapsulation preventing CML-034 access	2024-11-14	IMG-0847-034
P-031	CML-047	Inaccessible	Pipe rack Tier 3 — no scaffold access to CML-047	2024-11-15	IMG-0847-047
P-032	CML-048	Inaccessible	Active steam trace at CML-048 — temp too high	2024-11-15	IMG-0847-048
P-040	Calibration	Daily Cal	AM calibration setup, Gauge 08741, Nov 4	2024-11-04	IMG-0847-CAL-01
P-041	Calibration	Daily Cal	AM calibration setup, Gauge 11205, Nov 5	2024-11-05	IMG-0847-CAL-02

Complete photo log contains 247 photographs across all categories. Full resolution images available upon request from PNDE Records Management.

APPENDIX C — PERSONNEL CERTIFICATION RECORDS

Copies of current personnel certification records are maintained on file. The following summary confirms all personnel meet the requirements of GLCP-NDT-001 §3.1 and ASNT SNT-TC-1A.

Inspector	Cert Body	Method	Level	Cert No.	Issue Date	Expiry Date	Status
M. Rajendran	CGSB	UT	Level II	UT-2019-4472	2019-07-15	2025-07-15	Current
M. Rajendran	CGSB	MT	Level II	MT-2018-3201	2018-05-10	2024-05-10	Renewed
K. Nakamura	CGSB	UT	Level II	UT-2020-5891	2020-03-22	2026-03-22	Current
K. Nakamura	CGSB	PT	Level II	PT-2019-4105	2019-09-01	2025-09-01	Current
A. Thompson	CGSB	UT	Level II	UT-2021-6234	2021-09-01	2026-09-01	Current
S. Petrov	CGSB	UT	Level III	UT-2015-2108	2020-11-30	2025-11-30	Current
S. Petrov	CGSB	RT	Level III	RT-2016-2845	2021-06-15	2026-06-15	Current

Full certification documentation (scanned certificates) maintained in PNDE HR File Ref: HR-CERT-2024-0847. Available for client audit upon request.

APPENDIX D — EQUIPMENT CALIBRATION CERTIFICATES

Calibration certificates for all inspection equipment are issued by accredited calibration laboratories with traceability to NIST standards, per GLCP-NDT-001 §4.2. Certificate details are summarized below. Scanned originals are appended.

Equipment	Serial No.	Cal Lab	Cal Cert No.	Cal Date	Due Date	Traceable To
38DL PLUS Gauge	08741	MetCal Inc.	MC-2024-14521	2024-04-15	2025-04-15	NIST
38DL PLUS Gauge	11205	MetCal Inc.	MC-2024-18903	2024-08-22	2025-08-22	NIST
4-Step Cal Block	CB-0034	MetCal Inc.	MC-2023-09877	2023-06-10	2025-06-10	NRC/NIST

APPENDIX E — ISOMETRIC DRAWINGS WITH CML LOCATIONS

Isometric piping drawings with all CML locations clearly marked are maintained as controlled documents per API 570 §7 and API 574. The following drawing list identifies the isometrics applicable to the NHT-100 piping survey. CML locations are marked with a triangular symbol and sequential CML identification number.

Drawing No.	Title	Rev	Lines Covered	CML Range
GLCP-ISO-NHT-101	Naphtha Feed Line Isometric	4	NHT-L-101	CML-001 to CML-007
GLCP-ISO-NHT-102	Reactor Effluent Line Isometric	4	NHT-L-102	CML-008 to CML-015
GLCP-ISO-NHT-103	Stripper Overhead Line Isometric	3	NHT-L-103	CML-016 to CML-021
GLCP-ISO-NHT-104	H2 Makeup Line Isometric	4	NHT-L-104	CML-022 to CML-027
GLCP-ISO-NHT-105	Reactor Feed Line Isometric	3	NHT-L-105	CML-028 to CML-035
GLCP-ISO-NHT-106	Separator Outlet Line Isometric	4	NHT-L-106	CML-036 to CML-042
GLCP-ISO-NHT-107	Recycle Gas Line Isometric	3	NHT-L-107	CML-043 to CML-050
GLCP-ISO-NHT-108	Stripper Bottoms Line Isometric	3	NHT-L-108	CML-051 to CML-057
GLCP-ISO-NHT-109-114	Utility & Auxiliary Lines (Combined)	2	NHT-L-109 to L-114	CML-058 to CML-087

Marked-up isometric drawings are provided as separate PDF attachments (Files: GLCP-ISO-NHT-101 through GLCP-ISO-NHT-109-114). Digital originals maintained in GLCP Document Control system.

APPENDIX F — RAW DATA (GAUGE DATA EXPORTS)

Raw gauge data exports from the Olympus 38DL PLUS instruments are provided in electronic format (CSV files). The data files contain all individual thickness readings, including the four readings at 90-degree intervals (12, 3, 6, and 9 o'clock positions) for each CML. A summary of the raw data format is shown below.

Table F.1 — Sample Raw Data (5 of 82 CMLs shown; complete dataset in electronic files)

CML	Line	Position	Reading 1 (12 o'clock)	Reading 2 (3 o'clock)	Reading 3 (6 o'clock)	Reading 4 (9 o'clock)	Min	Date/Time	Oper
001	NHT-L-101	Str. Pipe	0.268	0.275	0.262	0.270	0.262	11-04 07:15	MR
002	NHT-L-101	Elbow	0.258	0.264	0.251	0.260	0.251	11-04 07:35	MR
003	NHT-L-101	Tee	0.262	0.270	0.258	0.265	0.258	11-04 08:00	MR
011	NHT-L-102	Tee	0.218	0.225	0.211	0.220	0.211	11-07 09:45	KN
024	NHT-L-104	Elbow	0.142	0.155	0.134	0.148	0.134	11-13 10:30	MR

Electronic raw data files:

- PNDE-GLCP-2024-0847-RAW-Gauge08741.csv (328 readings)
- PNDE-GLCP-2024-0847-RAW-Gauge11205.csv (224 readings)

- PNDE-GLCP-2024-0847-RAW-COMBINED.xlsx (master file with all readings)

— END OF REPORT —

This report and all appendices constitute the complete Final Inspection Report for the NHT-100 Naphtha Hydrotreater UT Thickness Survey.
Distribution is limited to authorized GLCP personnel per the distribution list maintained by GLCP Document Control.